

Foundation (Jalapeno)

Q1. Solve the equation $|x| = 5$.

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. What is the solution to $|2x| = 8$?

- (A) $x = 2$ or $x = -2$
- (B) $x = 4$ or $x = -4$
- (C) $x = 1$ or $x = -1$
- (D) $x = 0$
- (E) $x = 3$

Q3. A spaceship is traveling at a speed of $|v| = 200$ km/h. What are the possible values of v ?

- (A) $v = 100$ km/h
- (B) $v = 200$ km/h
- (C) $v = 200$ km/h or $v = -200$ km/h
- (D) $v = 0$ km/h
- (E) $v = 50$ km/h

Q4. Solve the equation $|x + 2| = 3$.

- (A) $x = 1$ or $x = -5$
- (B) $x = 2$ or $x = -4$
- (C) $x = 3$ or $x = -3$
- (D) $x = 0$
- (E) $x = 6$

Q5. A scientist is conducting an experiment with a particle that

has an absolute value of $|a| = 4$ m/s. What are the possible values of a ?

- (A) $a = 2$ m/s
- (B) $a = 4$ m/s
- (C) $a = 4$ m/s or $a = -4$ m/s
- (D) $a = 0$ m/s
- (E) $a = 8$ m/s

Q6. Solve the equation $|3x - 1| = 7$.

- (A) $x = 2$ or $x = -2$
- (B) $x = 3$ or $x = -1$
- (C) $x = 2.67$ or $x = -2$
- (D) $x = 0$
- (E) $x = 1$

Q7. A planet is orbiting a star with an absolute value of $|d| = 10$ AU. What are the possible values of d ?

- (A) $d = 5$ AU
- (B) $d = 10$ AU
- (C) $d = 10$ AU or $d = -10$ AU
- (D) $d = 0$ AU
- (E) $d = 20$ AU

Q8. Solve the equation $|2x + 5| = 9$.

- (A) $x = 2$ or $x = -7$
- (B) $x = 1$ or $x = -3$
- (C) $x = 2$ or $x = -9$
- (D) $x = 0$
- (E) $x = 4$

Open Ended Questions (FRQ)

Q9. Solve the equation $|x^2 - 4| = 5$ and check for extraneously solutions. Show all work and explain your reasoning.

Q10. A space station is experiencing a gravitational force with an absolute value of $|F| = 20$ N. If the force is given by the equation $|F| = |2x + 1|$, solve for x and check for extraneously solutions. Show all work and explain your reasoning.

Chef's Tips

- Check for extraneously solutions
- Use the definition of absolute value to solve equations
- Show all work and explain reasoning